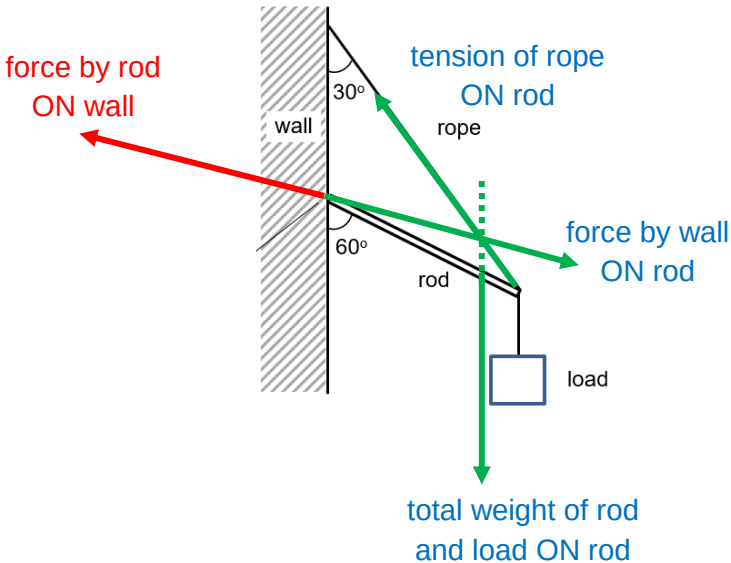
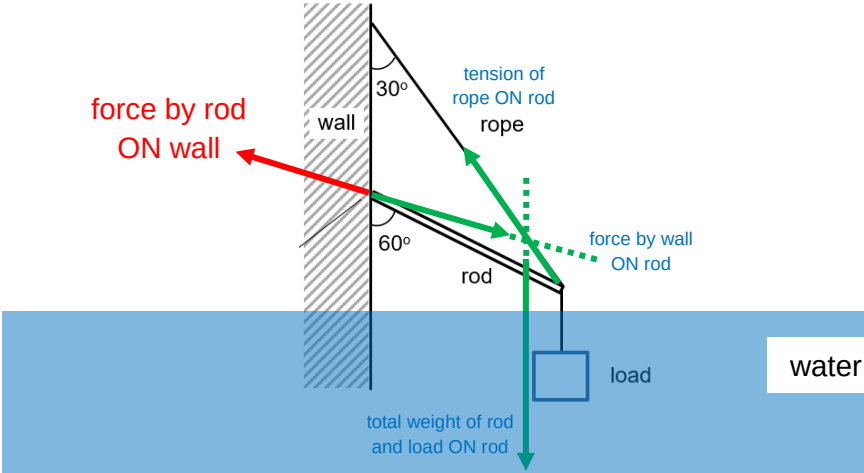


3	(a)		The moment of a force is equal to the <u>product of the force and the perpendicular distance</u> of the line of action of the force <u>from the pivot</u> . It is the turning effect of a force.	B1
	(b)	(i)	The diagram shows a horizontal beam pivoted to a vertical wall on the left. A downward force of 400 N is applied to the beam at a point where the beam makes a 60° angle with the wall. A support cable is attached to the end of the beam and the wall, making a 30° angle with the beam. A load of 2000 N is suspended from the end of the beam. The cable tension T is shown, along with its components $T \cos 30$ (vertical) and $T \sin 30$ (horizontal).	

			Take moments about the hinge, $(400)\left(\frac{L}{2} \sin 60\right) + 2000(L \sin 60) = T \sin 30(L)$ $T = 3810 \text{ N}$	B1 C1 A1
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		(ii)		
	(c)		 <u>The direction of all forces remain unchanged since the load remains submerged in water at the same location.</u> <u>The upthrust acting on the load will reduce the tension in the rope. Since the horizontal component of the tension reduces, the force by wall on rod also reduces. This will result in a lower value of force by rod on wall.</u>	B1 (direction) B1 (magnitude)

Questions	Answers	Marks
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4	(a)		kinetic energy was converted to gravitational potential energy and	R1
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